Mandana Samiei

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LinkedIn, GitHub

Education

Surgical Innovation Fellow in Experimental Surgery Program **Fall 2017- Summer 2018**
Industrial Group, working with Kinova Co., McGill University
We proposed a solution called AutoFeeder, to make the eating task by Jaco arm automatic not as individual movements. Our proposal was a motion tracking and path planning algorithm to find the optimal trajectory for feeding task.

M.Sc. in Computer Science **Fall 2017- present (Expected to graduate on Sept. 2019)**
Concordia University, Montreal, Quebec, Canada
Supervisor: Prof. Thomas Fevens
Courses: Applied Machine Learning, Image Processing, Surgical Innovation 1 & 2, Autonomous Vehicles (Duckietown)
GPA: A

B.Sc. in Computer Engineering **Fall 2012- Summer 2016**
Shahid Beheshti University (National University of Iran), Tehran, Iran
Thesis: Human Activity Recognition by finding patterns in LQI and RSSI signals generated by wireless sensors. Applying J48 decision tree and 10-fold cross validation using Weka software to classify the collected data including acceleration in x, y and z-axis, energy consumption and signal indicators. Supervisor: Prof. Mona Ghassemian
GPA: 17.22 / 20 Ranked 3rd among Computer Engineering Students Entered 2012

Research Experiences

Research Intern at Mila **Nov 2018- present**
Research Project: Providing a collection of TCGA benchmark clinical tasks that can be used in a meta-learning framework. Supervisor: Joseph Paul Cohen

Research Assistant at Dr. Fevens Lab. **Fall 2017- present**
Computer Science Department, Concordia University, Montreal, Canada
Research Scope: Medical imaging using novel deep learning architectures and assistive technologies

Research Assistant at Computer Network and Network Security (CNNS) Research Lab. **Winter 2016**
Shahid Beheshti University, Tehran, Iran, Advisor: Prof. Maghsoud Abbaspour
Research Topic: Analyzing Users' Behaviors in Twitter to Predict Social Bots
Developing a web application to retrieve users profile information to simulate the twitter environment using Twitter API and coded in Java programming language

Research Assistant at eSense Research Lab. **Summer 2015 – Summer 2016**
Shahid Beheshti University, Tehran, Iran, Advisor: Prof. Mona Ghassemian
Research Project: Human Activity Recognition (my bachelor thesis)

Professional Experiences

Research and Development Intern at THINK Surgical, Montreal, Canada **Summer 2018- present**
Job Description: Accuracy Assessment of Optical Tracking for Orthopedic Robotic Surgical Tool

Software Developer at DatX Co., Tehran, Iran (Internship Project) **Summer 2015**
Job Description: Working on a Telegram bot sending stock exchange information- Developed by PHP

Technical Committee Member at Smart Health and Well-being union **Winter 2015**
Supporting running the workshops tutorial sessions on setting up and programming sunspot sensors
Shahid Beheshti University, Tehran, Iran

Technical Skills

Programming Languages: Python, C++, MATLAB, PHP, HTML, CSS, JS, JSON, SQL

Tools: Pytorch, Keras, OpenCV, Jupyter Notebook, Pycharm, Qt Creator, NetBeans, Microsoft Visual Studio, PhpStorm, MySQL Workbench

Operating Systems: Macintash, Linux (Ubuntu), Windows

Publication

Accepted Mitacs Application **Summer, Fall 2018**
Title: Accuracy Assessment of Optical Tracking for Orthopedic Robotic Surgical Tool, IT Number: IT12789
Mitacs Research Project: In the proposed project, conducted an accuracy assessment of the multiple camera OTS used for the TNAV robotic surgical tool being developed by Think Surgical Inc. using a variety of metrics and analysis of these metrics to determine how stable the positional accuracy of the overall tracking system remains over time and under various lighting conditions.

Workshop Paper **Winter 2017**
P. Abed**, M. Samiei**, M. Ghassemian "Human Activity Recognition Deployment Using Wearable Wireless Sensors", **Accepted** in the IEEE Wireless Communications and Networking Conference **(**WCNC 2017).

Honors and Awards

Awarded Concordia Merit Scholarship **Summer 2018**
Concordia University, 10000 $CAD

Awarded a stipend as part of the NSERC-CREATE program **Fall 2017**
McGill University, 5000 $CAD for 1 years

Awarded Concordia Merit Entrance Scholarship **Fall 2017**
Concordia University, 5000 $CAD for 2 years

Ranked top 1% in the National University Entrance Examination (Konkoor) **Summer 2012**
Among more than 260,000 participants, Iran

Notable Projects

Meta-Learning and Genomics **Winter 2019**
Implemented Baselines (Logistic Regression, Majority, MLP and Prototypical Networks) using Pytorch and applied on TCGA dataset, Dealing with Class Imbalance in multi label classification tasks. Aims at creating a Benchmark of TCGA clinical task that can be used in a meta-learning regime.

An Ablation Study of Twin Networks: Matching the Future for Sequence Generation **Winter 2018**
Reproduce the Twin Net result on the Sequential MNIST
Final Project for Applied Machine Learning, McGill University

Kaggle Competition: Modified MNIST Challenge **Winter 2018**
Preprocessed images to make it similar to MNIST, Using Baselines algorithms (SVM, Logistic Regression), implemented NN from scratch, used Keras for CNN and tweaked hyper parameters, Course Project for Machine Learning, McGill University

Sentiment Classification of Yelp & IMDB dataset **Winter 2018**
Using Naïve Bayes', Decision tree and SVM, Compared the results and tuned hyper parameters of the models by cross validation, Binary/Frequency bag of words, Course Project for Machine Learning

Model Selection & Evaluation by using Cross-validation **Winter 2018**
Implemented gradient decent for regression as well as ridge/lasso regression, several hyperparameter tuning to find the best model. Course Project for Machine Learning, McGill University

Linear Classification & K-Nearest Neighbor **Winter 2018**
Implemented the algorithms and did a comparison between linear and non-linear classifiers, Created synthetic dataset with Gaussian distribution, Course Project for Machine Learning, McGill University

Binarization of degraded document images using Matlab. **Fall 2017**
Developed a Shannon-based Entropy binarization and a denoising method using morphological operators and shrink and swell filter. Classified pixels into 3 regions (text, near-text, non-text) and relabeled near-text pixels using Otsu thresholding algorithm. Course Project for Image Processing, Concordia University

Developed a Job-Finder Web Application (Bridge) using J2EE, Hibernate & Spring Framework
Working as a front-end and back-end developer, Designing Database using MySQL, UML and Workbench
Course Project for Internet Engineering, Shahid Beheshti University, Fall 2016

Simplified Design of a Chat Bot (Restaurant Finder) **Fall 2015**
The system is enabled to search between various types of restaurant and gives you suggestions based on your favorite foods. It mainly acts like a human, when you join the bot, it greets you and asks your favorites and recommends somewhere matched your interests. (As a recommender system)**.** Course Project for Artificial Intelligence, Shahid Beheshti University

Volunteer Experiences

Volunteering at NIPS 2018 **Fall 2018**
Volunteering at Montreal AI Symposium 2018 **Summer 2018**
Volunteering at RoboCup 2018 Montreal **Summer 2018**
Volunteering at McGill Career Fair **Winter 2018**

Executive Member in international workshop on smart health and wellbeing **Fall 2015**
Member of International Technical Committee **Spring 2015- Spring 2017**
Cooperated as an editor of scientific articles, Shahid Beheshti University, Tehran, Iran

Member of Computer Engineering Scientific Association **Fall 2012 - Summer 2013**
keeping track of Executive worksheet reports, Shahid Beheshti University, Tehran, Iran

Teaching Assistance Experiences

Fundamentals of Programming, Tutor/Marker**,** Concordia University, CA	**Winter 2019**
Fundamentals of Programming, Tutor/Marker**,** Concordia University, CA	**Fall 2018**
Principle of Programming Language, Programming on Duty, Concordia University, CA	**Winter 2018**
Computer Technical English, Head TA, Shahid Beheshti University, Tehran, Iran	**Winter 2017**
Computer Networks, Tutor, Shahid Beheshti University, Tehran, Iran	**Fall 2016**
Machine Language and Assembly, Marker**,** Shahid Beheshti University, Iran	**Fall/Spring 2015**